

# Gilbert Strang - A First Course in Calculus

## Chapters 1 - 7

### 1 Chapter 1: Introduction to Vectors

#### 1.1 Vectors and Linear Combinations

Problems 1-9 are about addition of vectors and linear combinations.

1. Describe geometrically (line, plane, or all of  $R^3$ ) all linear combinations of

(a)  $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$  and  $\begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$

(b)  $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$  and  $\begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$

(c)  $\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$  and  $\begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$  and  $\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$

2. Draw  $v = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$  and  $w = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$  and  $v + w$  and  $v - w$  in a single  $xy$  plane.

3. If  $v + w = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$  and  $v - w = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$ , compute and draw the vectors  $v$  and  $w$ .

4. From  $v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$  and  $w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ , find the components of  $3v + w$  and  $cv + dw$ .

5. Compute  $u + v + w$  and  $2u + 2v + w$ . How do you know  $u, v, w$  lie in a plane?

These lie in a plane because  
 $w = cu + dv$ . Find  $c$  and  $d$

$$u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad v = \begin{bmatrix} -3 \\ 1 \\ -2 \end{bmatrix}, \quad w = \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix}.$$

6. Every combination of  $v = (1, -2, 1)$  and  $w = (0, 1, -1)$  has components that add to \_\_\_\_\_.  
Find  $c$  and  $d$  so that  $cv + dw = (3, 3, -6)$ . Why is  $(3, 3, 6)$  impossible?

7. In the  $xy$  plane mark all nine of these linear combinations:

$$c \begin{bmatrix} 2 \\ 1 \end{bmatrix} + d \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{with} \quad c = 0, 1, 2 \quad \text{and} \quad d = 0, 1, 2.$$

8. The parallelogram in Figure 1.1 has diagonal  $v + w$ . What is its other diagonal? What is the sum of the two diagonals? Draw that vector sum.

9. If three corners of a parallelogram are  $(1, 1)$ ,  $(4, 2)$ , and  $(1, 3)$ , what are all three of the possible fourth corners? Draw two of them.

Problems 10–14 are about special vectors on cubes and clocks in Figure 1.4.

10. Which point of the cube is  $i + j$ ? Which point is the vector sum of  $i = (1, 0, 0)$  and  $j = (0, 1, 0)$  and  $k = (0, 0, 1)$ ? Describe all points  $(x, y, z)$  in the cube.
11. Four corners of this unit cube are  $(0, 0, 0)$ ,  $(1, 0, 0)$ ,  $(0, 1, 0)$ ,  $(0, 0, 1)$ . What are the other four corners? Find the coordinates of the center point of the cube. The center points of the six faces are \_\_\_\_\_. The cube has how many edges?
12. Review Question. In  $xyz$  space, where is the plane of all linear combinations of  $i = (1, 0, 0)$  and  $i + j = (1, 1, 0)$ ?

13. (a) What is the sum  $V$  of the twelve vectors that go from the center of a clock to the hours 1:00, 2:00, ..., 12:00?
- (b) If the 2:00 vector is removed, why do the 11 remaining vectors add to 8:00?
- (c) What are the  $x, y$  components of that 2:00 vector  $v = (\cos \theta, \sin \theta)$ ?
14. Suppose the twelve vectors start from 6:00 at the bottom instead of  $(0, 0)$  at the center. The vector to 12:00 is doubled to  $(0, 2)$ . The new twelve vectors add to \_\_\_\_\_.

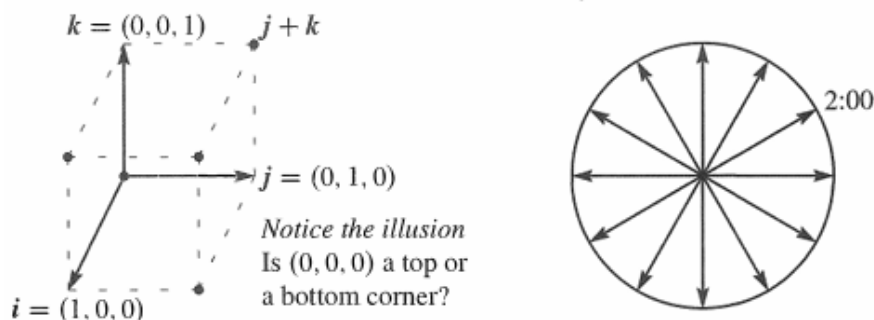


Figure 1.4: Unit cube from  $i, j, k$  and twelve clock vectors.

Problems 15–19 go further with linear combinations of  $v$  and  $w$  (Figure 1.5a).

15. Figure 1.5a shows  $\frac{1}{2}v + \frac{1}{2}w$ . Mark the points  $\frac{3}{4}v + \frac{1}{4}w$  and  $\frac{1}{4}v + \frac{3}{4}w$  and  $v + w$ .
16. Mark the point  $-v + 2w$  and any other combination  $cv + dw$  with  $c + d = 1$ . Draw the line of all combinations that have  $c + d = 1$ .
17. Locate  $\frac{1}{3}v + \frac{1}{3}w$  and  $\frac{2}{3}v + \frac{2}{3}w$ . The combinations  $cv + cw$  fill out what line?
18. Restricted by  $0 \leq c \leq 1$  and  $0 \leq d \leq 1$ , shade in all combinations  $cv + dw$ .
19. Restricted only by  $c \geq 0$  and  $d \geq 0$  draw the “cone” of all combinations  $cv + dw$ .

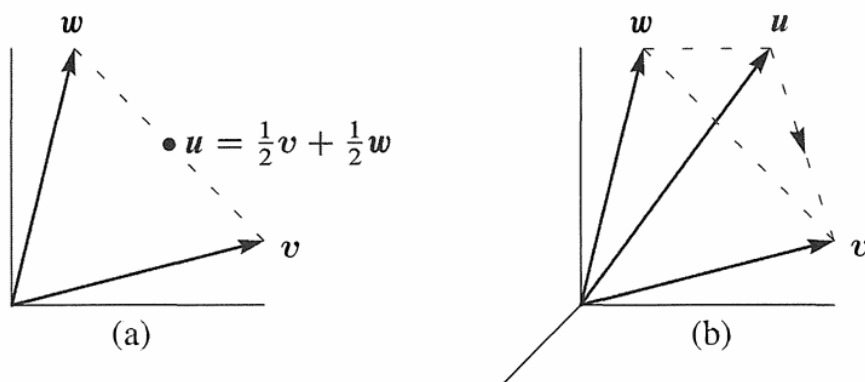


Figure 1.5: Problems 15-19 in a plane

Problems 20-25 in 3-dimensional space

Problems 20-25 deal with  $u, v, w$  in three-dimensional space (see Figure 1.5b).

20. Locate  $\frac{1}{2}v + \frac{1}{2}w + \frac{1}{2}v$  and  $\frac{1}{2}v + \frac{1}{2}w + \frac{1}{2}w$  in Figure 1.5b. Challenge problem: Under what restrictions on  $c, d, e$ , will the combinations  $cu + dv + ew$  fill in the dashed triangle? To stay in the triangle, one requirement is  $c \geq 0, d \geq 0, e \geq 0$ .
21. The three sides of the dashed triangle are  $v - u$  and  $w - v$  and  $u - w$ . Their sum is \_\_\_\_\_. Draw the head-to-tail addition around a plane triangle of  $(3, 1)$  plus  $(-1, 1)$  plus  $(-2, -2)$ .

22. Shade in the pyramid of combinations  $cu + dv + ew$  with  $c \geq 0, d \geq 0, e \geq 0$  and  $c + d + e \leq 1$ . Mark the vector  $\frac{1}{2}(u + v + w)$  as inside or outside this pyramid.
23. If you look at all combinations of those  $u, v$ , and  $w$ , is there any vector that can't be produced from  $cu + dv + ew$ ? Different answer if  $u, v, w$  are all in \_\_\_\_\_.
24. Which vectors are combinations of  $u$  and  $v$ , and also combinations of  $v$  and  $w$ ?
25. Draw vectors  $u, v, w$  so that their combinations  $cu + dv + ew$  fill only a line. Find vectors  $u, v, w$  so that their combinations  $cu + dv + ew$  fill only a plane.
26. What combination  $c \begin{bmatrix} 2 \\ 1 \end{bmatrix} + d \begin{bmatrix} 3 \\ 1 \end{bmatrix}$  produces  $\begin{bmatrix} 14 \\ 8 \end{bmatrix}$ ? Express this question as two equations for the coefficients  $c$  and  $d$  in the linear combination.

### Challenge Problems

27. How many corners does a cube have in 4 dimensions? How many 3D faces? How many edges? A typical corner is  $(0, 0, 1, 0)$ . A typical edge goes to  $(0, 1, 0, 0)$ .
28. Find vectors  $v$  and  $w$  so that  $v + w = (4, 5, 6)$  and  $v - w = (2, 5, 8)$ . This is a question with \_\_\_\_\_ unknown numbers, and an equal number of equations to find those numbers.
29. Find two different combinations of the three vectors  $u = (1, 3)$  and  $v = (2, 7)$  and  $w = (1, 5)$  that produce  $b = (0, 1)$ . Slightly delicate question: If I take any three vectors  $u, v, w$  in the plane, will there always be two different combinations that produce  $b = (0, 1)$ ?
30. The linear combinations of  $v = (a, b)$  and  $w = (c, d)$  fill the plane unless \_\_\_\_\_.
31. Find four vectors  $u, v, w, z$  with four components each so that their combinations  $cu + dv + ew + fz$  produce all vectors  $(b_1, b_2, b_3, b_4)$  in four-dimensional space.
32. Write down three equations for  $c, d, e$  so that  $cu + dv + ew = b$ . Can you somehow find  $c, d, e$  for  $b$ ?

$$u = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \quad v = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

## 1.2 Lengths and Dot Products

### Challenge Problems

## 1.3 Matrices

### Challenge Problems

## **2 Chapter 2: Solving Linear Equations**

### **2.1 Vectors and Linear Equations**

Challenge Problems

### **2.2 The Idea of Elimination**

Challenge Problems

### **2.3 Elimination Using Matrices**

Challenge Problems

### **2.4 Rules for Matrix Operations**

Challenge Problems

### **2.5 Inverse Matrices**

Challenge Problems

### **2.6 Elimination = Factorization: $A = LU$**

Challenge Problems

### **2.7 Transposes and Permutations**

Challenge Problems

### **3 Chapter 3: Vector Spaces and Subspaces**

#### **3.1 Spaces of Vectors**

Challenge Problems

#### **3.2 The Nullspace of $A$ : Solving $Ax = 0$ and $Rx = 0$**

Challenge Problems

#### **3.3 The Complete Solution to $Ax = b$**

Challenge Problems

#### **3.4 Independence, Basis and Dimension**

Challenge Problems

#### **3.5 Dimensions of the Four Subspaces**

Challenge Problems

## **4 Chapter 4: Orthogonality**

### **4.1 Orthogonality of the Four Subspaces**

Challenge Problems

### **4.2 Projections**

Challenge Problems

### **4.3 Least Squares Approximations**

Challenge Problems

### **4.4 Orthonormal Bases and Gram-Schmidt**

Challenge Problems

## **5 Chapter 5: Determinants**

### **5.1 The Properties of Determinants**

Challenge Problems

### **5.2 Permutations and Cofactors**

Challenge Problems

### **5.3 Cramer's Rule, Inverses, and Volumes**

Challenge Problems

## **6 Chapter 6: Eigenvalues and Eigenvectors**

### **6.1 Introduction to Eigenvalues**

Challenge Problems

### **6.2 Diagonalizing a Matrix**

Challenge Problems

### **6.3 Systems of Differential Equations**

Challenge Problems

### **6.4 Symmetric Matrices**

Challenge Problems

### **6.5 Positive Definite Matrices**

Challenge Problems



## **7 Chapter 7: The Singular Value Decomposition (SVD)**

### **7.1 Image Processing by Linear Algebra**

**Challenge Problems**

### **7.2 Bases and Matrices in the SVD**

**Challenge Problems**

### **7.3 Principal Component Analysis (PCA by the SVD)**

**Challenge Problems**

### **7.4 The Geometry of the SVD**

**Challenge Problems**